



Multimodal LLMs for Historical Document Analysis

W3 - OCR, Post-Correction, and NER in Historical Documents

AI for Programmers (CSO – 432)

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The Challenge: Digitizing Historical Archives

Problem & Motivation

Noisy Data

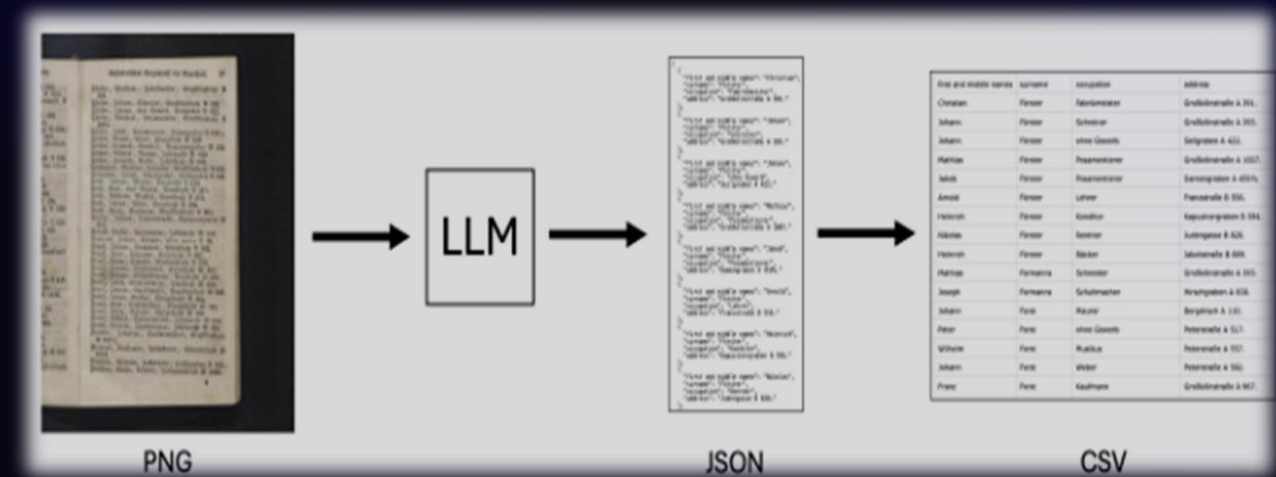
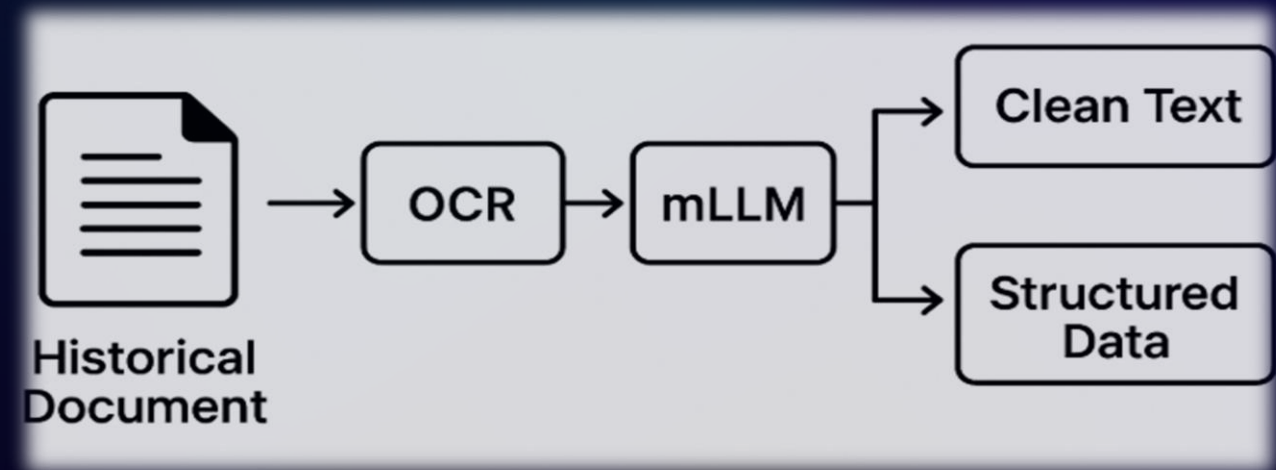
Vast archives of digitized Fraktur/Gothic documents (census, trade records) are time-consuming and error-prone to process.

OCR Limitations

Conventional OCR struggles with noisy scans, old fonts, and layout variability, requiring hours of manual cleaning by historians.

Real-World Impact

Critical for digital humanities, economic history, and genealogy; enables searchable databases and preserves cultural heritage.



Novel Approach: An mLLM-Based Pipeline

Existing methods are limited by character-level recognition and brittle, rule-based post-processing. We propose an integrated solution.



OCR + Post-Correction

Accurate text extraction and correction using visual and linguistic context.



Entity Recognition (NER)

Identifying and structuring key data points (names, occupations, addresses).



Direct CSV Output

Producing high-quality, ready-to-analyze structured datasets (CSV/JSON).

Integration includes Gemini-2.0 Flash, LLaVA, Florence-2, and Tesseract. Achieved **77% exact** and **96% fuzzy accuracy** on benchmark datasets.

Gaps in Related Work

We bridge the gap between visual understanding and strict, schema-constrained structured extraction for historical Fraktur text.

End-to-End OCR Systems <i>(2018–2023)</i>	CNN / Transformer models	Weak on Fraktur/degraded scans; poor layout handling.
Neural Post-Correction <i>(2019–2024)</i>	Seq2Seq models	Misses visually ambiguous errors; depends heavily on initial OCR quality.
Layout-Aware Transformers <i>(2020–2024)</i>	LayoutLM, DocFormer	Needs dense annotations; fails on historical/irregular layouts.
Multimodal LLMs <i>(2023–2025)</i>	GPT-4V, Gemini	Hallucination risk; poor schema control for structured output.

📌 Our contribution: A reproducible pipeline merging visual understanding with schema-constrained extraction, robustly handling Fraktur text and outputting CSV-ready records deterministically.

Proposed Method: End-to-End Architecture

The system converts historical document images into structured CSV datasets in a single automated pipeline.



Core Innovation

- **Unified Architecture:** Bridges OCR and structured data generation.
- **Prompt-Driven Reasoning:** Employs multimodal reasoning to replace manual steps (OCR, correction, tagging).
- **Output-Aware Design:** Ensures consistent, schema-constrained CSV formatting for downstream analysis.

Task Definition and Model Objective

Goal: Image to Structured Table

Automatically convert a historical document image into a structured CSV table containing:

- First & Middle Names
- Surname
- Occupation
- Address
- Page Number

Each row represents one person or record extracted from the document.

Model Functionality (One Step)



Read Text (OCR)



Correct Errors (Post-correction)



Identify & Organize Entities (NER + Structuring)

Evaluation Metrics: Exact vs. Fuzzy Accuracy

We use a combination of strict and tolerant matching to provide a realistic measure of performance, accounting for historical orthographic variations.



Exact Match Accuracy

Measures character-for-character correctness (case-insensitive) between predicted and ground-truth text.

$$\text{ExactAcc} = \frac{\text{Exact Matches}}{\text{Total Entries}} \times 100$$



Fuzzy Match Accuracy

Uses Jaro-Winkler similarity (threshold 0.90) to allow small spelling or accent differences (e.g., "Müller" vs "Mueller").

$$\text{FuzzyAcc} = \frac{\text{Fuzzy Matches } (\geq 0.90)}{\text{Total Entries}} \times 100$$



Overall Accuracy (Weighted)

Combines strict and tolerant matching to evaluate the overall extraction quality and human-level readability.

$$\text{OverallAcc} = \frac{\text{ExactAcc} + \text{FuzzyAcc}}{2}$$

Experimental Setup and Data

Dataset & Preprocessing

- **Source:** Ground-truth benchmark from Griesshaber et al. (2025).
- **Data Type:** Historical German economic and city directory documents (Fraktur / Gothic print).
- **Input:** 10 multi-page PDFs, converted to PNGs at 300 DPI.
- **Preprocessing:** Enhanced contrast/sharpness; cleaned text (ligatures, special characters).

Figure 1: The First Page of each Directory in our Dataset



Implementation Details



Frameworks

Python 3.10, PyTorch, Hugging Face, Google Generative AI.



Models

Gemini 2.0 Flash, LLaVA, Florence-2, Tesseract (deu, deu_frak).



Setting

Zero-shot inference, Temperature 0.0 (deterministic generation).

Results and Discussion

Multimodal models show improved contextual reasoning, but structured output remains challenging.

Key Findings

- **Gemini 2.0 Flash** gave the best structured extraction accuracy (~96% fuzzy match and 77% exact match).
- **Traditional OCR (Tesseract)** struggles with historical documents and layout irregularities.
- **Preprocessing** and **structured prompt design** were critical factors, improving fuzzy accuracy by ~4%.
- Gemini correctly identifies and separates fields (name, occupation, address), indicating **semantic understanding**.

model	pdf	exact_match	fuzzy_match	rows_predicted	rows_ground_truth
gemini-2.0	type-1	77.04%	96.17%	98	98

First 3 rows:

id	first and middle names	surname	occupation	address	page_number
1	Christian	Förster	Fabrikmeister	Großkölnstraße A 391	1
2	Johann	Förster	Schreiner	Großkölnstraße A 393	1
3	Johann	Förster	ohne Gewerb	Seilgraben A 422	1

Failure Cases Analysis

Column Overrun

Model outputs 8+ fields instead of 6, causing CSV parsing failure and data loss.

OCR Noise

Fraktur characters misread (e.g., "f" as "ff"), leading to incorrect names or addresses.

Formatting Leakage

LLMs occasionally add markdown fences ("`csv"), interrupting the parsing process.

Conclusion and Future Vision

Hybrid OCR-LLM pipelines bridge the gap between visual archives and structured digital data, advancing digital humanities.

Scaling & Fine-Tuning

Fine-tune mLLMs on historical German datasets and integrate layout-aware models (LayoutLMv3).

Vision

Towards a robust, domain-adapted OCR-LLM system that can digitize and structure entire historical archives automatically and reliably.



Pipeline Enhancements

Add automatic language/font detection and adaptive CSV parsing for variable column outputs.

Deployment

Build a web-based interface for interactive processing and integrate cloud-based GPUs for speed.

References

Baseline Paper

Griesshaber, N. et al. (2024).

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